Session 1
The General Problem of Intelligence: Credit Assignment

Last session |

- AI and organizations = the same thing
- Intelligence = ability to learn to solve complex problems

Last session |

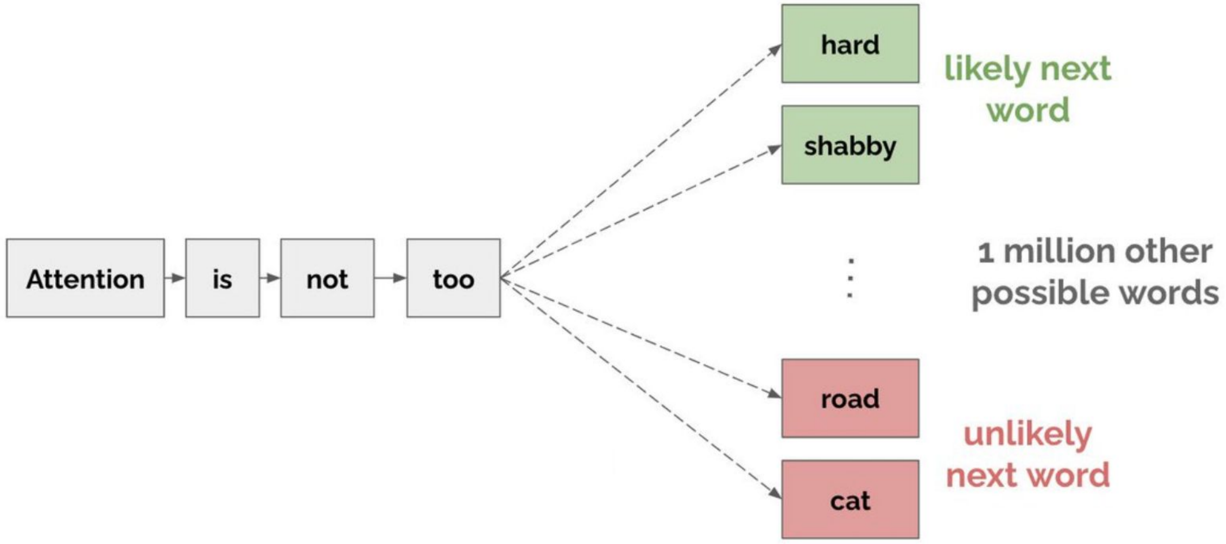
- AI and organizations = the same thing
- Intelligence = ability to <u>learn</u> to solve <u>complex problems</u>

- *We need a general way to talk about <u>what we actually mean</u> by "learning" and "complex" problems across diverse AI ideas and organizations*

This session |

- AI and organizations = the same thing
- Intelligence = ability to **learn** to solve complex problems
    - **Credit assignment** as the general problem of intelligent learning in AI and organizations

Before we begin! Currently-popular AI = specific to a “prediction machine” view of intelligent learning



Source:
https://threadreaderapp.com/thread/1481767733972901888

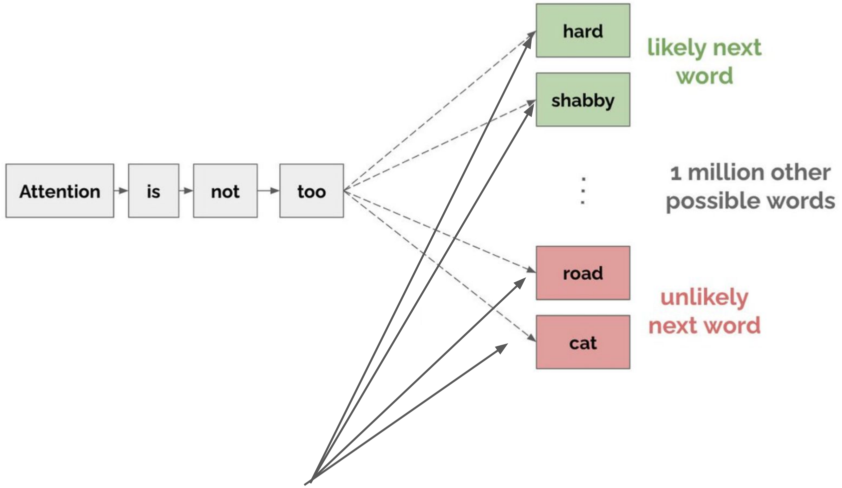
Most AI ideas ≠ machine learning ≠ “prediction machines”

- We need a more general way than prediction to talk about intelligent learning

Credit assignment

- Any process for **evaluating** (assigning credit to) **effects of individual actions** on solving a problem

Prediction is an <u>example</u> of credit assignment (e.g., of evaluating individual actions when solving a problem)



Individual actions = **Generating text fragments**

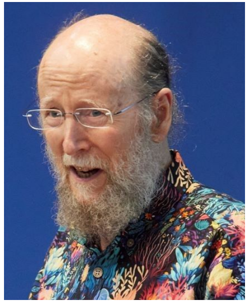
Credit assignment = **evaluating** whether the fragments are good

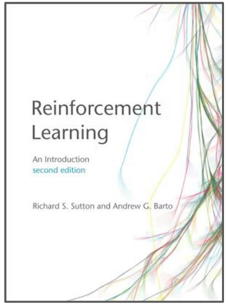
Source: https://threadreaderapp.com/thread/1481767733972901888

Let us look at credit assignment broadly, through two paradigms

- Reinforcement learning

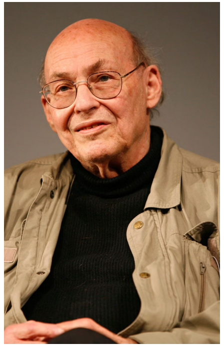

Arthur Samuel (IBM)


Richard Sutton (U. Alberta)

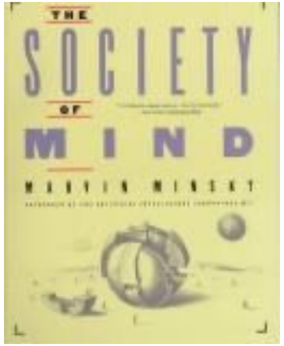


Sutton & Barto (book)

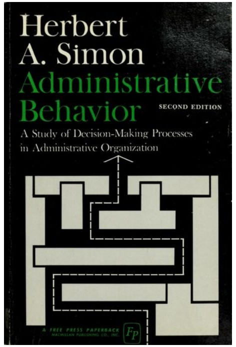
- Learning in a society


Marvin Minsky (MIT)


Oliver Selfridge (MIT)



Society of Mind (book) (1988), Minsky

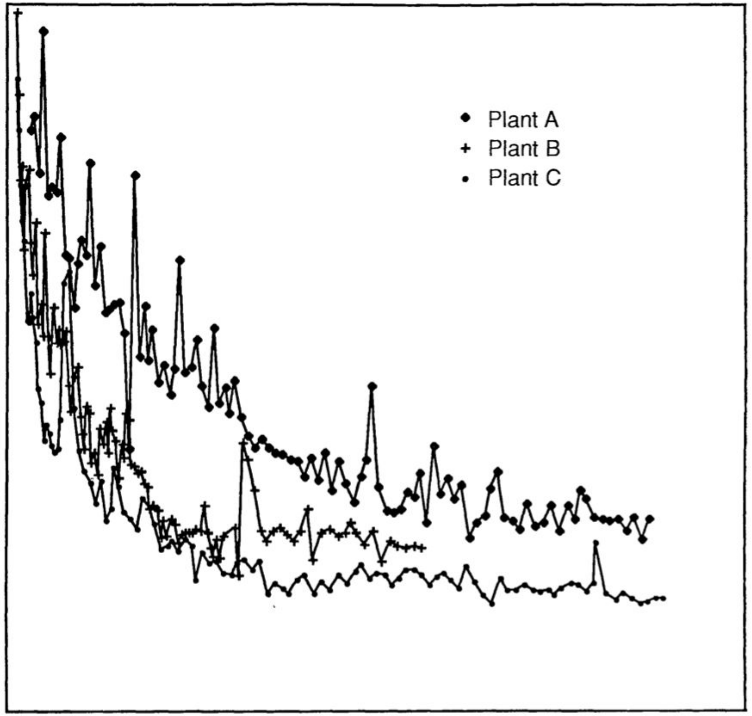


Admin. Behavior (book) (1947), Simon

Sources:
https://caiogasparine.medium.com/machine-learning-98e83b0d346f
https://news.nus.edu.sg/experience-beats-knowledge-prof-richard-sutton-on-reinforcement-learning-and-the-future-of-ai/
https://mitpress.mit.edu/9780262039246/reinforcement-learning/
https://www.independent.co.uk/news/people/marvin-minsky-dead-the-world-has-lost-one-of-its-greatest-minds-in-science-a6834981.html
https://www.computer.org/csdl/magazine/ex/1996/05/x5015/13rRUwbs2cJ
https://en.wikipedia.org/wiki/Society_of_Mind
https://archive.org/details/administrativebe00simo

(1) Shipbuilders learning to manufacture cost-efficiently

Cost per ship



# of ships produced

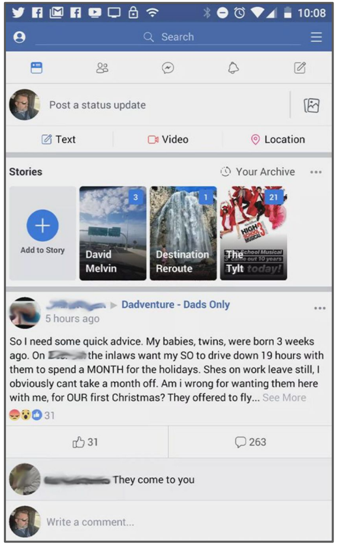
Costs go down from org.’s “**learning curve**” (not just economies of scale)

(Argote & Epple 1990)

Source:
https://www.science.org/doi/pdf/10.1126/science.247.4945.920?casa_token=lAioOJ3x5zcAAAAA:8fJzUzdt3Mg2Nsdn2owJoqC6FGIvnmoiiccw-mR9ue3TsQ5k8BmJxPDOIH4sAyoh7iZ2IGqLQp5aDQ

(2) An app dev. team learning the app features users want

How did Facebook "learn" to get over here



Facebook, 2004

Facebook, 2026

Source: https://en.wikipedia.org/wiki/Facebook
https://www.lifewire.com/best-facebook-apps-for-android-8774320

(3) Pharma firms learning to choose marketing budgets

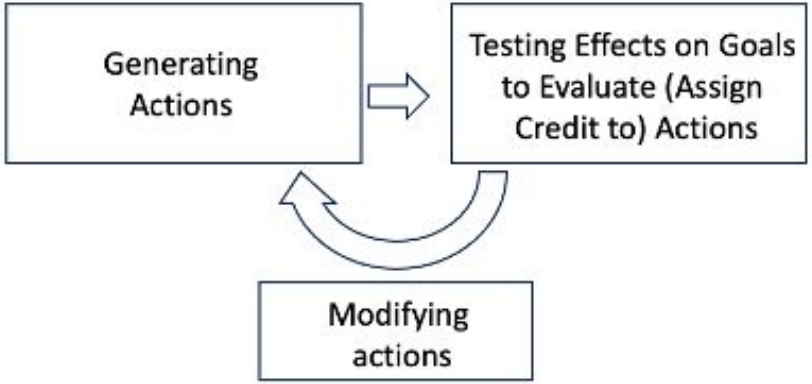
Why do firms in the same industry (e.g., pharmaceuticals) have different marketing budgets?

Learning from (different) experiences is one explanation.

Source: [executives committee meeting]

What to take away from the examples? Credit assignment = a process of *generating*, *testing*, and *modifying* actions

- Generating actions | *e.g., building ships, trying app features, allocating budget*
- Testing actions | *e.g., data on ship defects, user feedback, sales figures*
- Modifying actions | *e.g., refining shipbuilding techniques, adding or removing features, changing the budget*



Why was assigning credit challenging in the examples?

- Actions were **interdependent** (evaluating individual actions may depend on their effects on other actions)

- Feedback (ship defects, users' reactions to features, etc.) was **delayed** or **ambiguous**.

Arthur Samuel (1959)'s assign credit based on **intermediate feedback** about problems

Checkers example: count the pieces lost to evaluate your actions so far.

Source: https://theapothecarynh.com/?w=72255857081730
https://www.youtube.com/watch?v=5vGZRQu5HpE

General solution to assign credit = **intermediate feedback**

- Shipbuilding company: counting defects in parts
- App development team: defining key performance indicators (KPIs) for each step in a “customer journey”
- Budgeting team: milestones for each business unit

Two paradigms in AI for generating intermediate feedback to assign credit

- Arthur Samuel's (1959) approach at IBM, now called **reinforcement learning** (Sutton & Barto 2018).

- Marvin Minsky (1961) at MIT: Samuel's approach not general enough. He proposed an alternative that we will call **learning in a "society"** (Minsky 1988).

Paradigm 1 | Assigning credit by reinforcement learning

Reinforcement basics

- Agents assign **probabilities,** called “**beliefs**”, to **actions**
- Good actions get **rewarded** (= positive feedback)
- Agents assign credit by **modifying probabilities** (“reinforcing”) actions that get rewarded or not
- Overall strategy: learn to reinforce actions to maximize rewards

Reinforcement basics (continued)

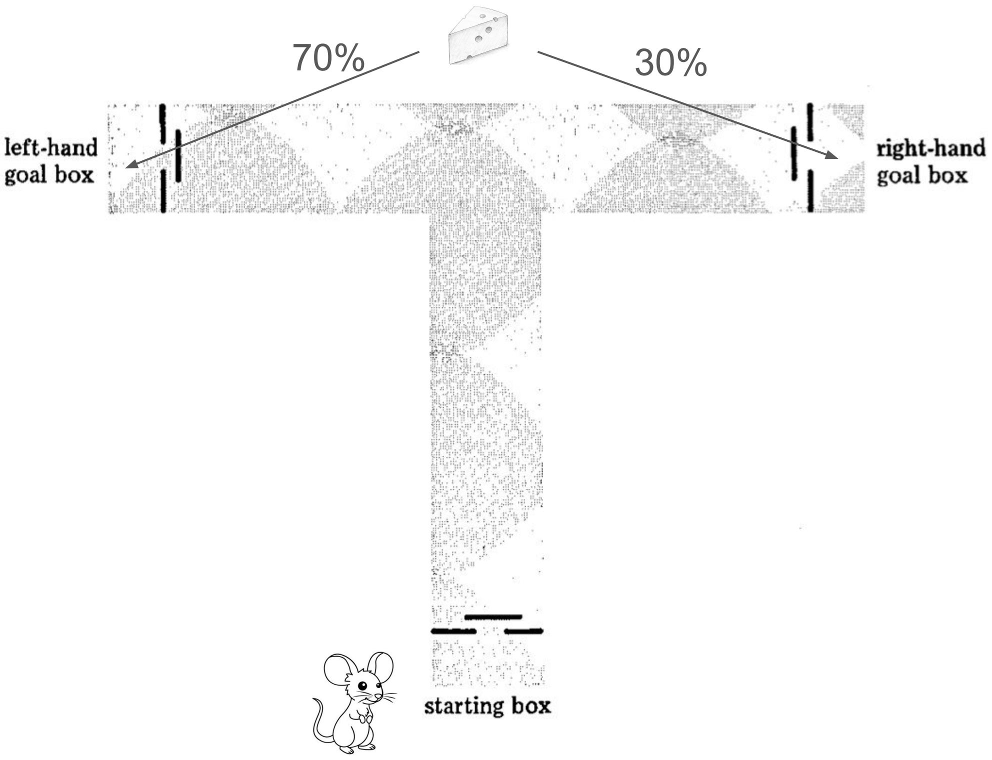
- Agents need a **policy** for how much credit to assign from feedback (how much to modify beliefs when rewarded)
- The basic policy is called the **learning rate**, which can differ for rewarded versus non-rewarded actions, and looks like this:

Updated belief = (*learning rate*$_{\text{success}}$)*(1 - *prior probability of generating action*)

or:

Updated belief = (*learning rate*$_{\text{failure}}$)*(1 - *prior probability of generating action*)

Navigating a T-Maze



- A mouse navigates a T-maze (the simplest kind of maze) 1,000 times to find cheese.
- Each trial, cheese is on the left w/70% probability, but never on the right.
- The mouse at first has no idea where the cheese is — it believes there's a 50–50 probability it's left or right.
- To maximize its cheese, the mouse needs to "learn" to turn left every time.

Source: Lave, C., March, J. 1975. An Introduction to Models in the Social Sciences. Ch. 6: Adaptation

T-Mazes — policies, rewards, and learning rates

TRIAL NO.	P_R AT BEGINNING OF THAT TRIAL	LEARNING INCREMENT
1	0.500	0.3 (1 — .500) = 0.150
2	0.650	0.3 (1 — .650) = 0.105
3	0.755	0.3 (1 — .755) = 0.074
4	0.829	0.3 (1 — .829) = 0.051
5	0.880	0.3 (1 — .880) = 0.036
6	0.916	0.3 (1 — .916) = 0.025
7	0.941	0.3 (1 — .941) = 0.018
8	0.959	0.3 (1 — .959) = 0.012
9	0.971	0.3 (1 — .971) = 0.009
10	0.980	

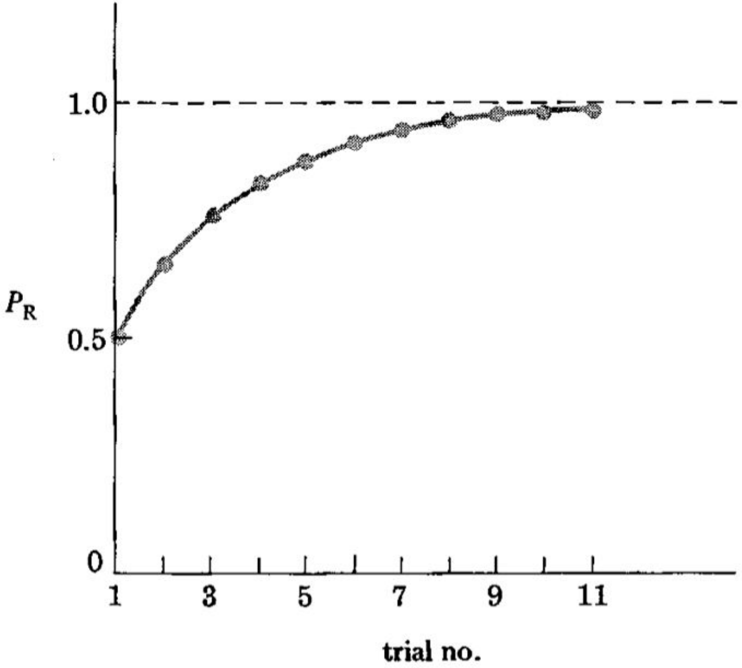
Policy |
Mouse's beliefs (P_R) = probabilities for going left or right.

Learning rate/increment |
How quickly/slowly to change beliefs based on feedback

Reinforcement |
Idea that "rewarded" behavior (finding the cheese) in one trial leads to similar behavior the next trial

Source: Lave, C., March, J. 1975. An Introduction to Models in the Social Sciences. Ch. 6: Adaptation

T-Mazes - Learning behavior



- The mouse “learns” over 11 trials, using a fast “learning rate”, the optimal choice of going right 100% of the time.

Source: Lave, C., March, J. 1975. An Introduction to Models in the Social Sciences. Ch. 6: Adaptation

The exploration-exploitation tradeoff

Trial	Probability of going left under a 10% learning rate (more "exploitative")	Probability of going left under a 30% learning rate (more "explorative")
1	0.5	0.5
2	0.55	0.65
3	0.595	0.755
4	0.636	0.829
5	0.682	0.880
6	0.714	0.916
7	0.742	0.941
8	0.768	0.959

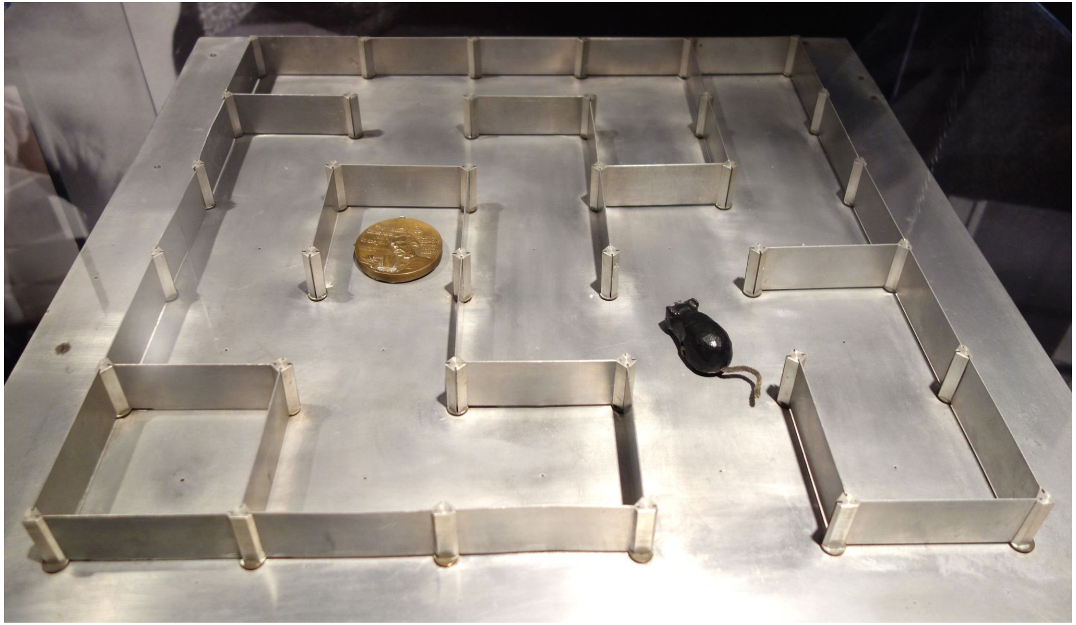
Look at the two columns: both mice will eventually "learn" to go left 100% of the time. Are the both equally intelligent? Not really: each trial has a **cost**. We want an **efficient** agent.

Higher learning rate: more "exploratory", faster, but risks learning the wrong things.

Lower learning rate: more "exploitative", safer, but risks learning too slowly.

Credit assignment solution: **balance** exploring and exploiting.

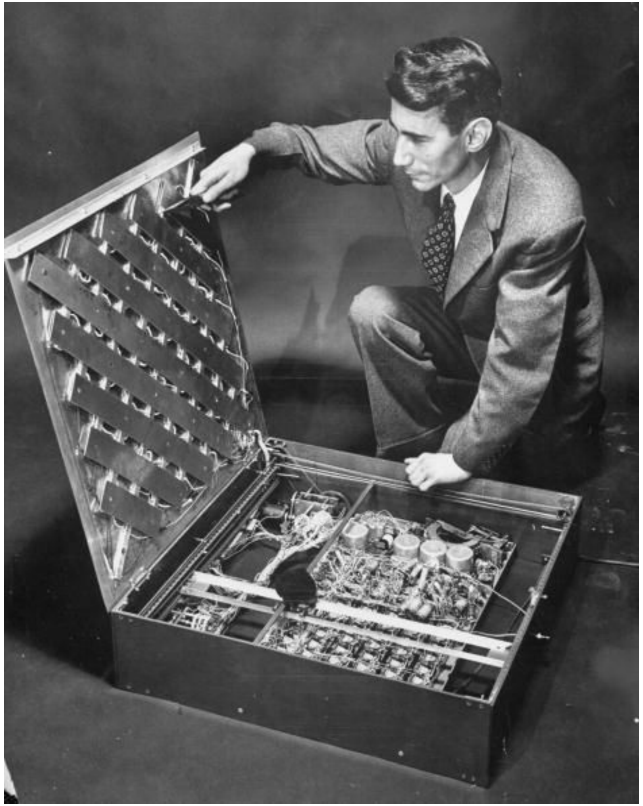
What if we add interdependent actions to our maze?



- A mouse tries to find the cheese but needs to make multiple turns.
- How is the learning problem different versus the T-maze?

Source:
https://commons.wikimedia.org/wiki/File:Theseus_Maze_by_Claude_Shannon,_1952_-_MIT_Museum_-_DSC03702.JPG

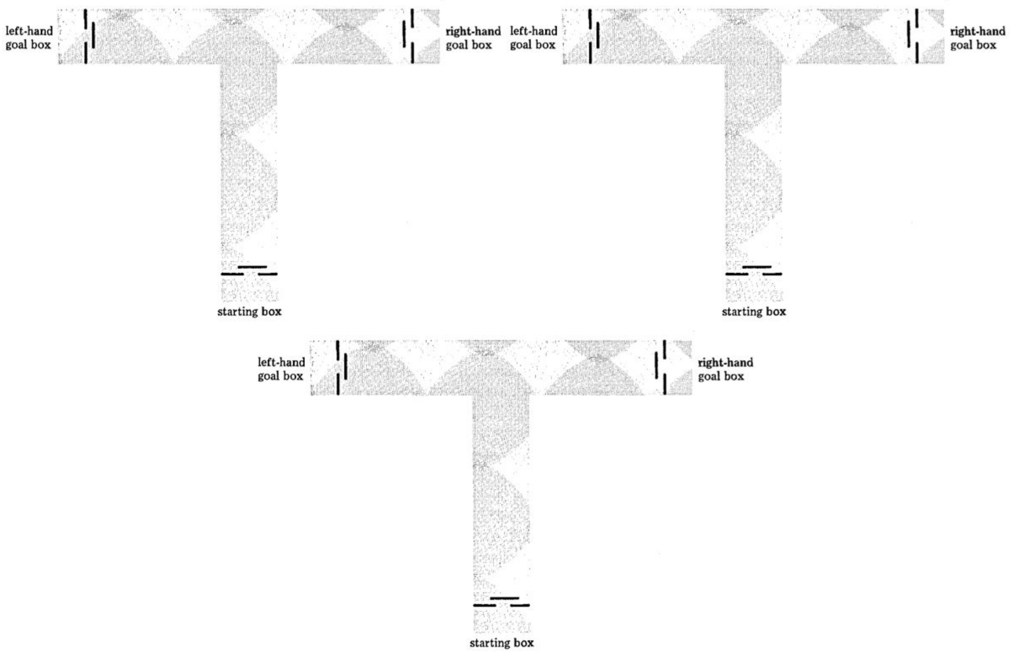
Theseus (ca. 1950)



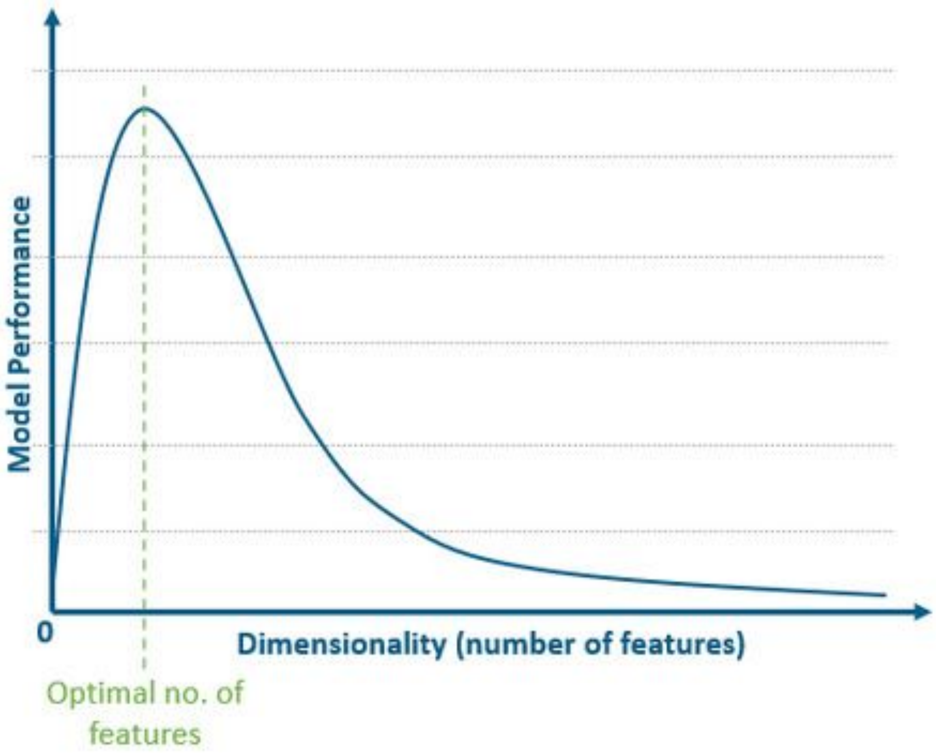
This labyrinth is called Theseus, created by Claude Shannon around 1950 at MIT

Source:
https://cyberneticzoo.com/mazesolvers/1952-%E2%80%93-theseus-maze-solving-mouse-%E2%80%93-claude-shannon-american/

Navigating T-mazes with two interdependent actions



“Curse of dimensionality” = hard to assign credit
(coined by Richard Bellman, 1950s — an operations mgmt. researcher)



As dimensions (e.g., **actions**) become larger, performance (i.e., learning ability) goes down **exponentially**.

This is also called “**combinatorial explosion**”

Credit assignment when navigating T-mazes with two interdependent actions

- **Combinatorial complexity** | A linear increase in interdependent actions leads to a combinatorial increase in the cost of learning to assign credit
- **Combinatorial explosion** | Even a small increase in interdependent actions quickly overwhelms the ability to assign credit by reinforcing beliefs about actions
- **Superstitious learning** | modifying actions based on beliefs ("superstitions") reinforced without adequate feedback to assign credit

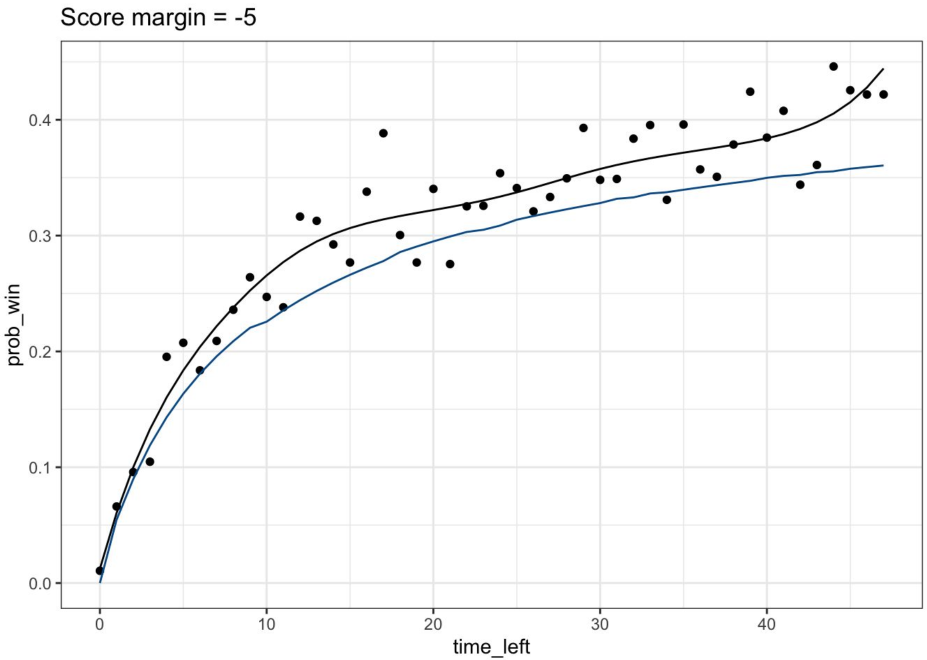
Assigning credit under combinatorial explosion using Sutton's **temporal differencing (TD)** approach

- Assign credit not to beliefs about actions.
- Instead: assign credit to a simpler "**value function**" — a function that estimates "**cumulative expected rewards**".

Example 1: chances of a team winning an NBA basketball game if it is currently losing by 5 points.

- “Cumulative expected value” (chances of winning) depends on how much time is left

- No beliefs about individual actions needed, just measuring “rewards” (chances of winning)

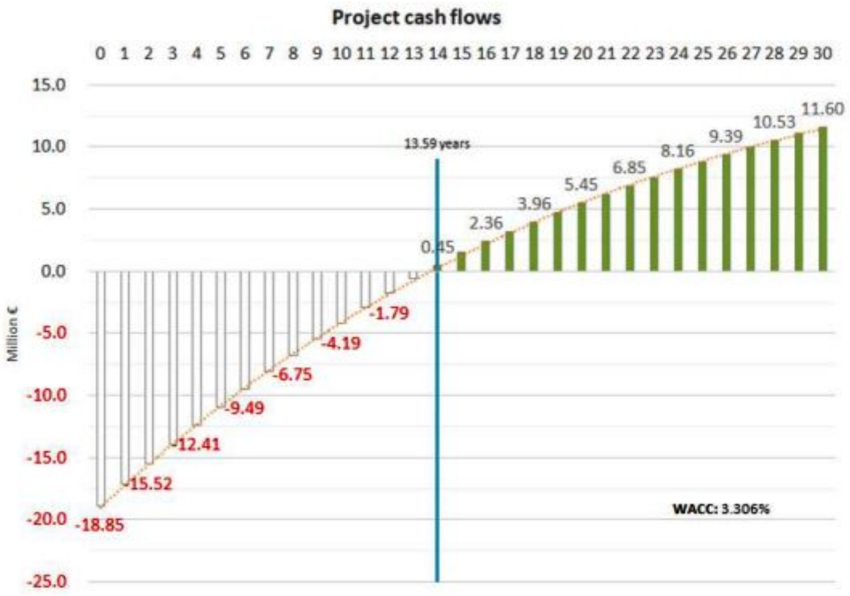


Source: https://blog.albertkuo.me/post/probability-of-winning-an-nba-game-a-minute-by-minute-breakdown/

Example 2: forecasting the value of a long-run investment by calculating “net present value” (NPV))

- “Cumulative expected value” (sum of future cash flows) depends on cash flows for each of 30 time periods.

- No beliefs about individual actions needed, just measuring actual “rewards” (cash flows) for any time period versus expected rewards.



Source: https://solarthermalworld.org/news/npv-irr-and-ddp-language-bankers-and-investors/

Temporal differencing (TD) (Sutton & Barto): use current feedback to update estimates of future feedback

- Key idea: can update the value function for any intermediate feedback received. Don’t have to wait for delayed rewards.

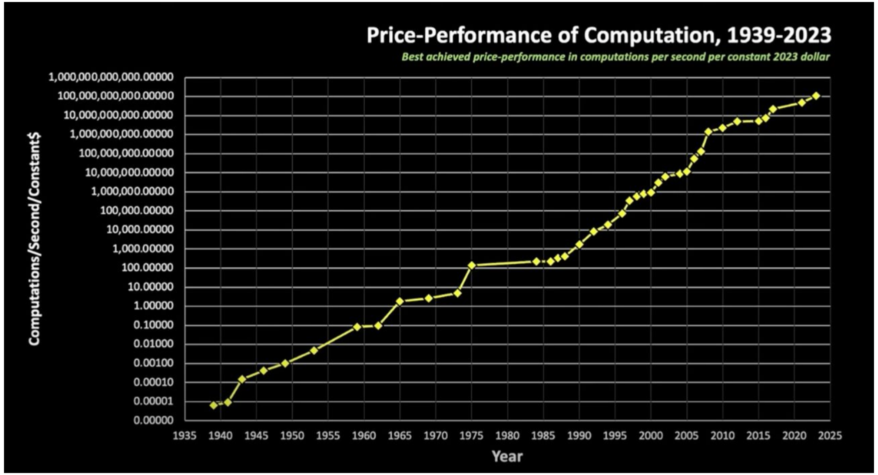
- Example: when investing in a project, managers use early cash flows to revise estimates for future cash flows without needing to know the final cash flows for the project.

Sutton's "reward hypothesis" as a general solution to the problem of credit assignment

*"All of what we mean by **goals** and purposes can be well thought of as the **maximization** of the expected value of the **cumulative sum** of a received scalar signal (called **reward**)"*

(Sutton & Barto 2018, p. 57)

What about **combinatorial explosion**? Sutton's "reward hypothesis" states that TD strategies for reinforcement learning should work with **big enough computers**.



2/3/22, 9:31 PM The Bitter Lesson

The Bitter Lesson

Rich Sutton

March 13, 2019

The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin. The ultimate reason for this is Moore's law, or rather its generalization of continued exponentially falling cost per unit of computation. Most AI research has been conducted as if the computation available to the agent were constant (in which case leveraging human knowledge would be one of the only ways to improve performance) but, over a slightly longer time than a typical research project, massively more computation inevitably becomes available. Seeking an improvement that makes a difference in the shorter term, researchers seek to leverage their human knowledge of the domain, but the only thing that matters in the long run is the leveraging of computation. These two need not run counter to each other, but in practice they tend to. Time spent on one is time not spent on the other. There are psychological commitments to investment in one approach or the other. And the human-knowledge approach tends to complicate methods in ways that make them less suited to taking advantage of general methods leveraging computation. There were many examples of AI researchers' belated learning of this bitter lesson, and it is instructive to review some of the most prominent.

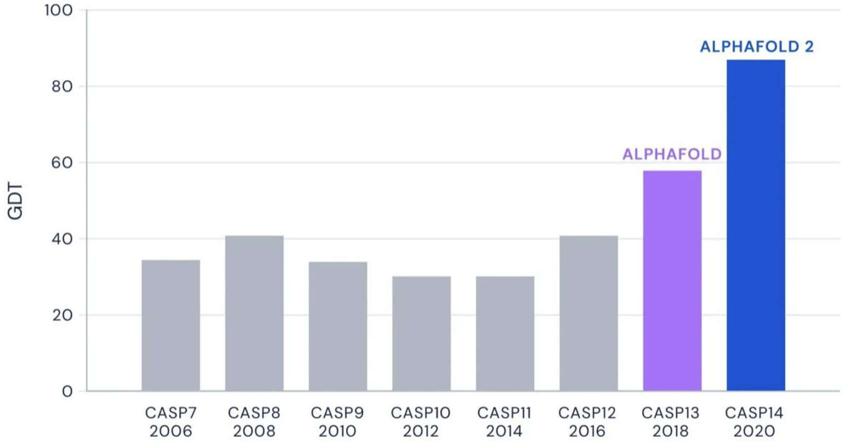
In computer chess, the methods that defeated the world champion, Kasparov, in 1997, were based on massive, deep search. At the time, this was looked upon with dismay by the majority of computer-

Source: https://www.bvp.com/atlas/ai-escape-velocity-a-conversation-with-ray-kurzweil
http://www.incompleteideas.net/IncIdeas/BitterLesson.html

(2a.) AlphaGo (2016): learning to play the game of Go

Defeated Go grandmaster Lee Sedol 4 games to 1: **reinforcement learning** (briefly) becomes wildly popular

(2b.) AlphaFold (2018,20,24): learning to predict protein structure



Goes from predicting 56% of protein structures to 87%, with similar techniques as AlphaGo

Source:
https://www.techniques-ingenieur.fr/actualite/articles/alphago-li-a-deep-mind-de-google-gagne-premier-match-de-go-32473/
https://deepmind.google/blog/alphafold-a-solution-to-a-50-year-old-grand-challenge-in-biology/

Is the “reward hypothesis” really a general solution to credit assignment?

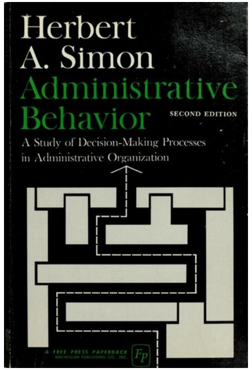
…Is reinforcement learning really the right paradigm to think about intelligence in AI and organizations?

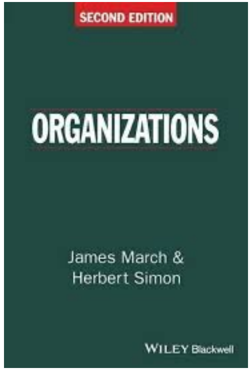
“Carnegie tradition” (1950s-present) in AI and org. theory: intelligence requires **complex decision-making processes**, not just reinforcement

- Go: combinatorially complex, but clear **given goal**
- Our earlier org. examples: *vastly more complex, uncertain, ambiguous*! Managers must plan, coordinate, evaluate multiple goals, etc.
- Real-world org. problems require **beliefs** about actions and goals. TD may help, but not fundamental.

Herbert Simon, Carnegie-Mellon

James March, Carnegie-Mellon/Stanford

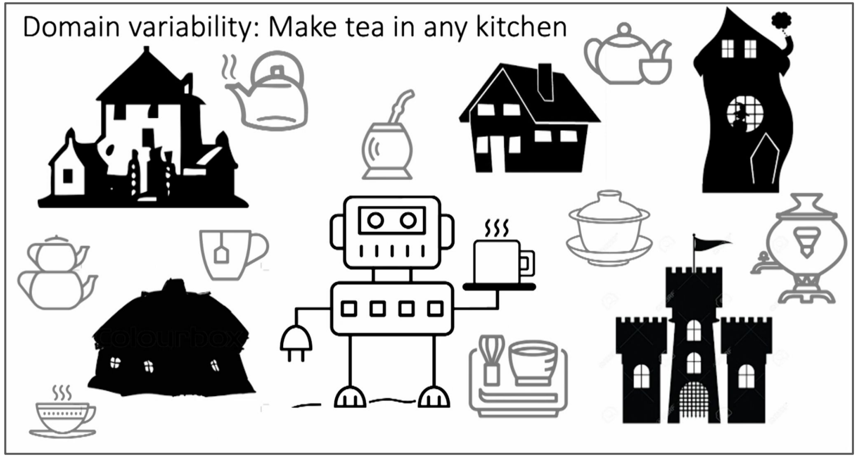




Sources: https://en.wikipedia.org/wiki/Herbert_A._Simon
https://www.gsb.stanford.edu/newsroom/school-news/james-g-march-professor-business-education-humanities-dies-90
https://archive.org/details/administrativebe00simo
https://product.kyobobook.co.kr/detail/S000010643697

MIT tradition in AI (1950s-present) | Problems often only **partially observable**, cannot reinforce unknown rewards

To make tea, we need to assign credit to **plans** a **sequence of actions** by **reasoning about objects** (kettles, water, cups, etc.) and about **causal structure** (what happens if water is poured, etc.).



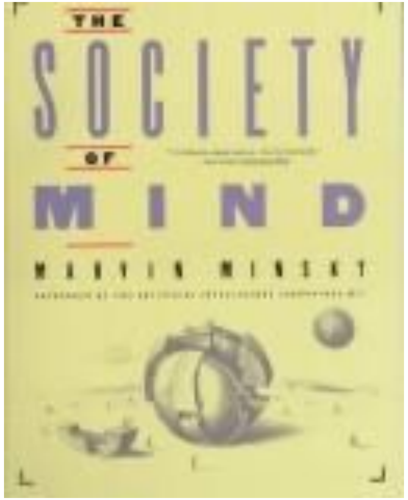
Leslie Kaelbling, MIT

Source: https://www.youtube.com/watch?v=mrB7Vs4CzLQ
https://www.eecs.mit.edu/people/leslie-kaelbling/

“You first must be able to do something before you can be rewarded for it”

… this requires understanding a problem, not just reinforcing rewards for a given problem.

Marvin Minsky
(MIT)

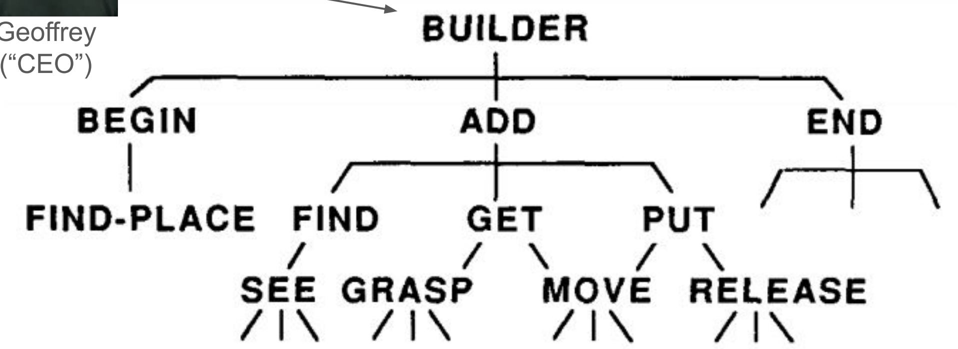


Society of Mind
(book) (1988)

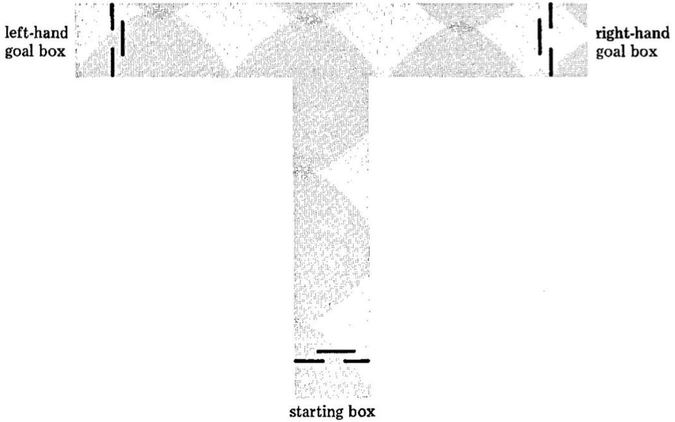
Paradigm 1 | Assigning credit by learning in a “society”

Example: before the mouse can navigate the T-maze, our CEO “Geoffrey” has to “build” it

Geoffrey
(“CEO”)



Agents in a Bureaucracy

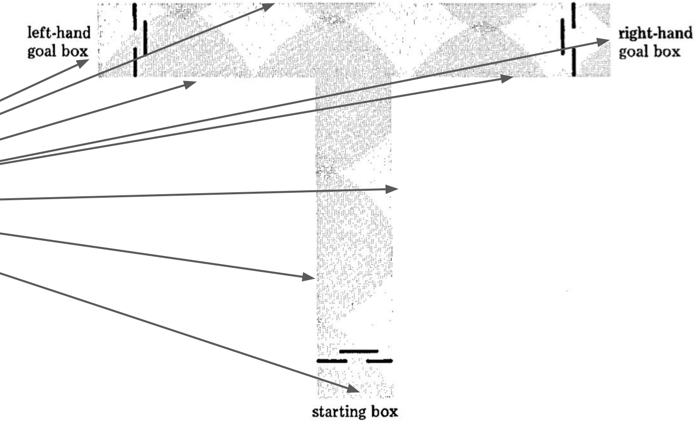


Source: https://x.com/mathemagic1an/status/1645096289770811397

One "reward" needed is "containment" (building the maze so that the mouse cannot escape)

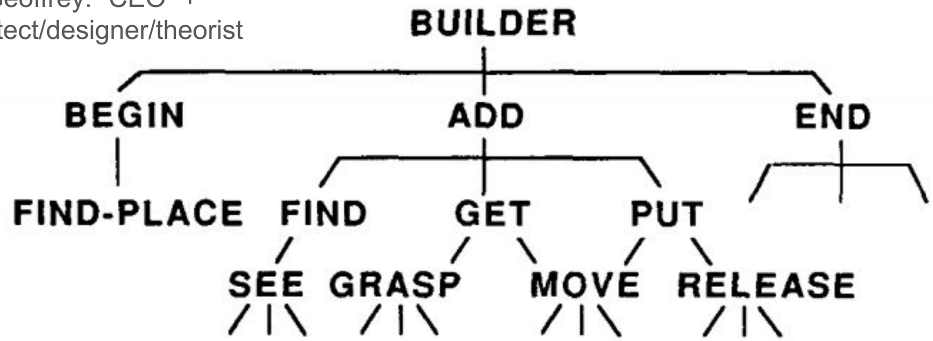
Yet "containment" only arises when all eight sections of the maze are built…

So *how to assign credit to individual actions*?

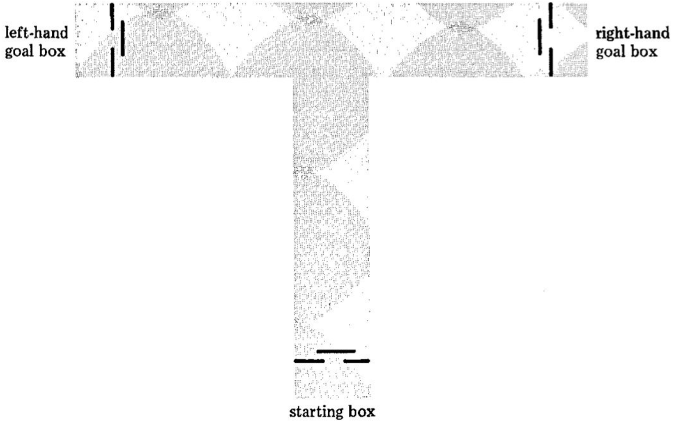


Credit is assigned *to the society's actions*… which implies that Geoffrey already has some theory/idea of "containment"

Geoffrey: "CEO" + architect/designer/theorist



Agents in a Bureaucracy



Source: https://x.com/mathemagic1an/status/1645096289770811397

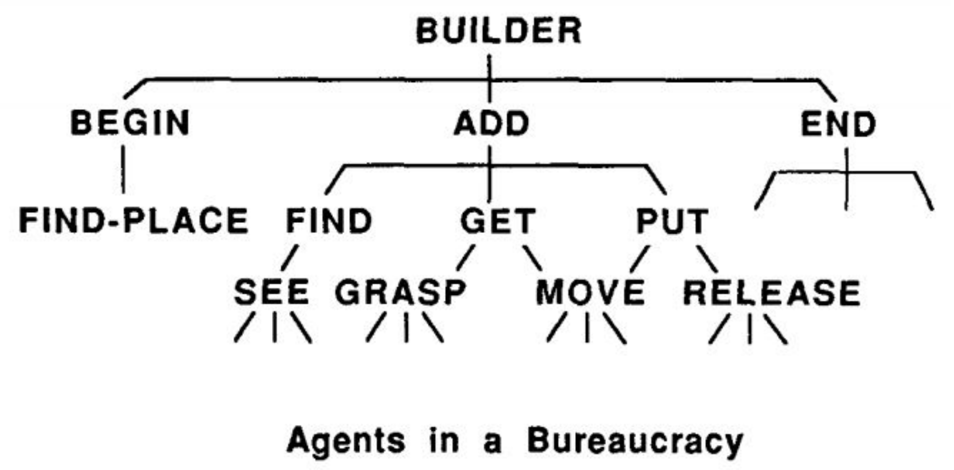
Learning in a society: basics

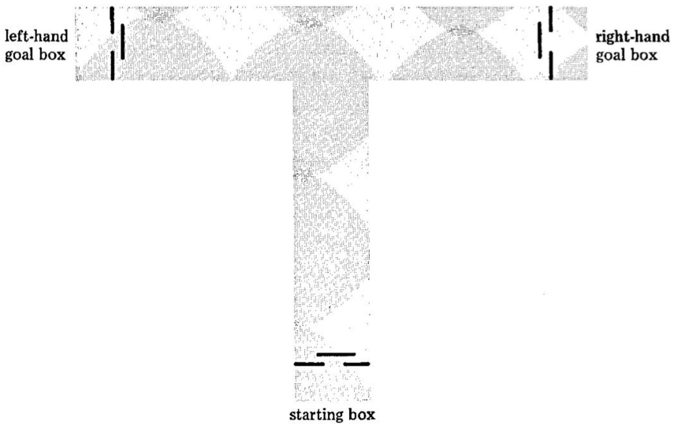
- (1) Complex problems = **sparse feedback** (only a small subset of actions generates feedback useful for assigning credit).
- (2) Intermediate feedback requires **understanding the problem**.
- (3) Problems are too complex to be understood by one agent.
- (4) Strategy: learn to assign credit as a **society of agents**.
- (5) Assigning credit requires some **theories/ideas/knowledge** about the structure of the complex problem.

Learning in a society: basics (continued)

- Agents learn **delegation actions** for managing other agents: in which situations they should be “recruited” or not.
- The “policy” is not a value function, but a mapping from situations to sets of agents.
- Learning = “reorganizing” agents by assigning credit locally to many agents at once, rather than a single reward.

Example: before the mouse can navigate the T-maze, our CEO "Geoffrey" has to build it



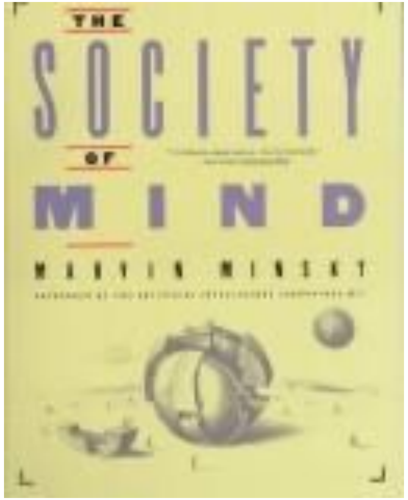


Source: https://x.com/mathemagic1an/status/1645096289770811397

“You first must be able to do something before you can be rewarded for it”

… this requires **managing** a society of agents based on ***decomposing a*** *problem into subproblems in some* ***meaningful way***.

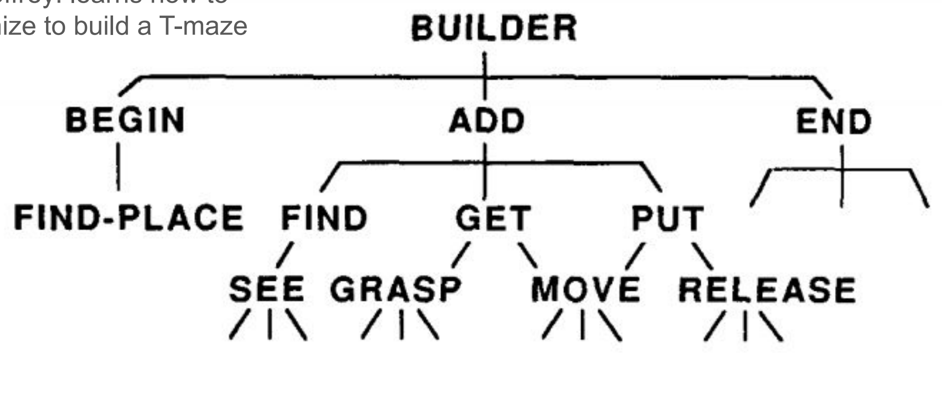
Marvin Minsky
(MIT)



Society of Mind
(book) (1988)

“Learning” in a society = not “exploring” or “exploiting” a value function, so much as understanding the functional meaning (e.g., containment) of different decompositions of problems and “societies” of agents

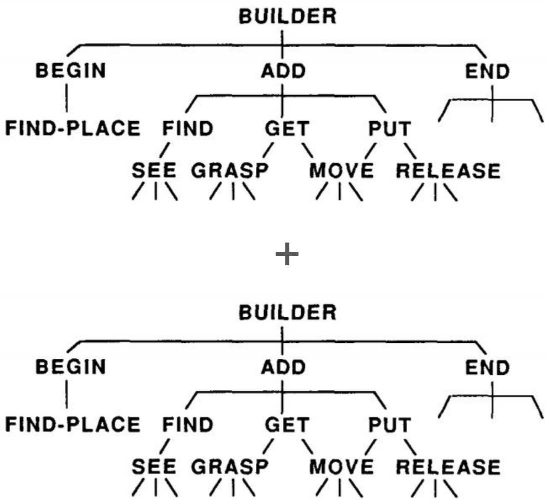
Geoffrey: learns how to organize to build a T-maze

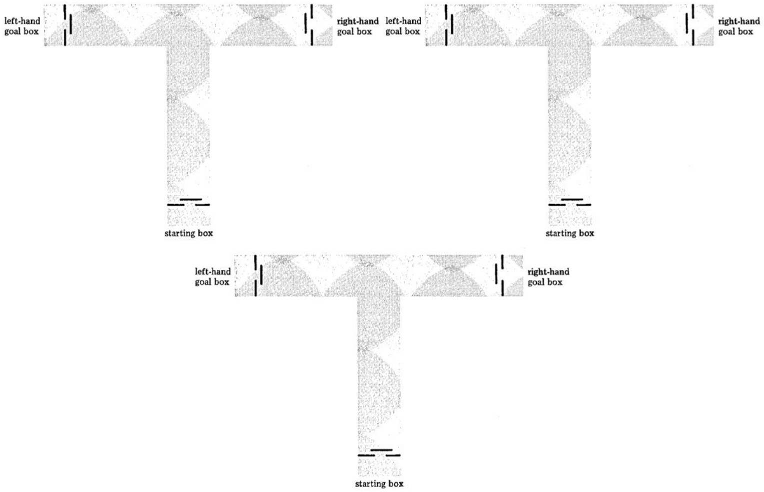


Agents in a Bureaucracy

Source: https://x.com/mathemagic1an/status/1645096289770811397

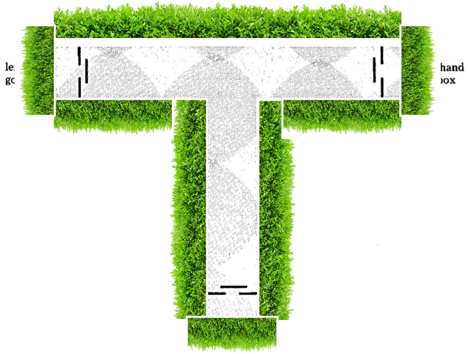
Building T-mazes with two interdependent actions: Geoffrey can just use the same society! Increase in complexity is <u>linear</u>, not combinatorial as in our mouse example.





Building two types of T-mazes: basic + one with hedges. Increase in complexity is <u>far greater</u> than combinatorial.

Now, Geoffrey must modify the society of agents itself!



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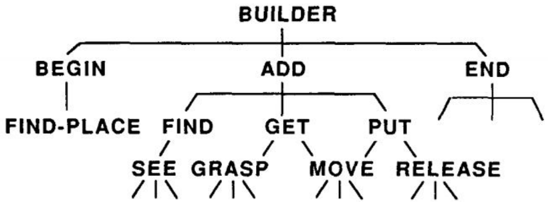

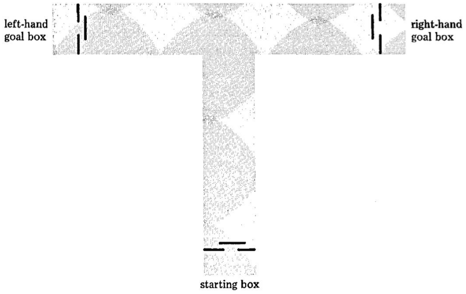

Here, intelligence requires learning to assign credit to some “**society of memories**” (i.e., figuring out what to remember)

CEO, architect/designer, and **organizer/manager of a system of knowledge/memory**

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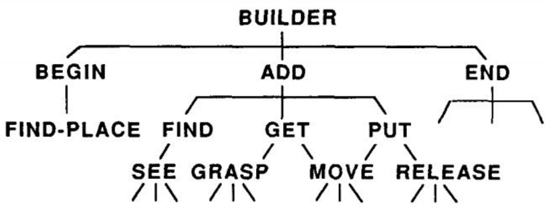

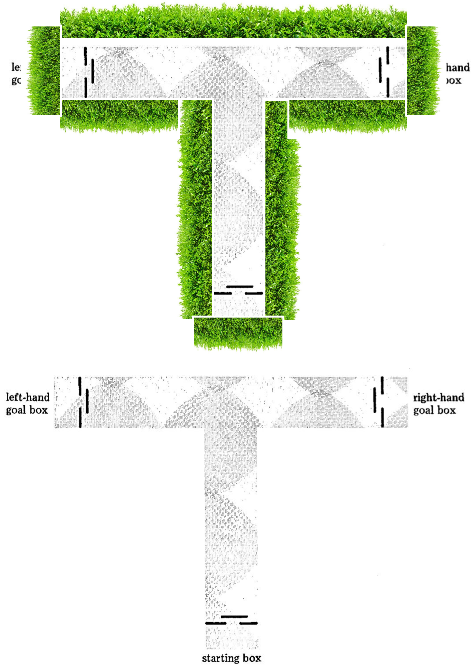

Minsky’s “management hypothesis” as a general solution to the problem of credit assignment

*All of what we mean by **goals** and purposes can be well thought of as **managing** processes of decomposing problems, and memories or knowledge about these problems, into societies of agents, and assigning credit within that society in ways that enable it to satisfy some higher-level evaluative functions.*

Returning to our organizational examples

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- **(3) The biotech firm hears of a new biotech manufacturing technology that will bring 10x productivity improvements. Which business units should it allocate R&D budget to, given that the results of R&D will not be seen for years?**

Review | Intelligence in AI and organizations in this course = learning to assign credit to actions

- **Credit assignment** = evaluating effects of actions on solving a problem
- **Process** of assigning credit = generating, testing, modifying actions
- AI has **contrasting paradigms** for assigning credit
  - Learning by reinforcing beliefs or value functions to maximize a reward
  - Learning by a society of agents to understand a problem and remember its essence
- **This course** will be about exploring a **variety of strategies for assigning credit**, both within and beyond these first two paradigms.